# CANONICAL EQUIVALENCE OF FUNCTION FIELDS OF TRANSCENDENCE DEGREE 1 WITH CURVES UNDER DOMINANT RATIONAL MAPS AND WITH NORMAL PROJECTIVE CURVES UNDER MORPHISMS

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the restriction of the varieties-level equivalence, the regular-normal-projective dictionary, the fully faithful functor with its two extension lemmas, the normal projective model construction, and the author's closing remark on the alternative compactify-then-normalize order are all retained verbatim.

*Editorial note.* The label theorem-curves-rational-maps names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own theorem environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Rational maps of varieties and function fields. Varieties, Theorem varieties-theorem-varieties-rational-maps with Lemma varieties-lemma-dimension-locally-algebraic. Supplies the equivalence between (1) and (2) by restriction: a variety is a curve exactly when its function field has transcendence degree 1.
- Extension of morphisms from the generic point. Same chapter, Lemmas lemma-extend-over-normal-curve and lemma-extend-over-dvr; Varieties, Lemmas varieties-lemma-dim-1-proper-projective, varieties-lemma-normalization-locally-algebraic, varieties-lemma-reduced-dim-1-projective-completion. Extends a rational map from a normal curve to a morphism, and produces the normal projective model of a given function field by normalizing and completing.

*Source reference.* Original reference 0BXY: sources/stacks-curves/pinned-original-source.tex, source lines 272–272. Prerequisite pointer invoked at proof line(s) 421; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0BXZ: sources/stacks-curves/pinned-original-source.tex, source lines 287–287. Prerequisite pointer invoked at proof line(s) 418; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0BXB: sources/stacks-chow/tags.txt, source lines 14974–14974. Prerequisite pointer invoked at proof line(s) 444; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original definition 02IS: sources/stacks-chow/tags.txt, source lines 3311–3311. Prerequisite pointer invoked at proof line(s) 400; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0BX2: sources/stacks-chow/tags.txt, source lines 14965–14965. Prerequisite pointer invoked at proof line(s) 404; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0569: sources/stacks-chow/tags.txt, source lines 6702–6702. Prerequisite pointer invoked at proof line(s) 403; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0BXV: sources/stacks-chow/tags.txt, source lines 14993–14993. Prerequisite pointer invoked at proof line(s) 440; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0A26: sources/stacks-chow/tags.txt, source lines 12694–12694. Prerequisite pointer invoked at proof line(s) 408; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0A21: sources/stacks-chow/tags.txt, source lines 12689–12689. Prerequisite pointer invoked at proof line(s) 395; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0BXR: sources/stacks-chow/tags.txt, source lines 14989–14989. Prerequisite pointer invoked at proof line(s) 432; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0BXW: sources/stacks-chow/tags.txt, source lines 14994–14994. Prerequisite pointer invoked at proof line(s) 435; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0B8Y: sources/stacks-chow/tags.txt, source lines 14156–14156. Prerequisite pointer invoked at proof line(s) 405; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original theorem 0BXN: sources/stacks-chow/tags.txt, source lines 14986–14986. Prerequisite pointer invoked at proof line(s) 392; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Theorem 0.1.** *Let $k$ be a field. The following categories are canonically equivalent*

(1) *The category of finitely generated field extensions $K/k$ of transcendence degree* 1*.*
(2) *The category of curves and dominant rational maps.*

(3) *The category of normal projective curves and nonconstant morphisms.*
(4) *The category of nonsingular projective curves and nonconstant morphisms.*
(5) *The category of regular projective curves and nonconstant morphisms.*
(6) *The category of normal proper curves and nonconstant morphisms.*

*Proof.* The equivalence between categories (1) and (2) is the restriction of the equivalence of Varieties, Theorem 0BXN. Namely, a variety is a curve if and only if its function field has transcendence degree 1, see for example Varieties, Lemma 0A21.

The categories in (3), (4), (5), and (6) are the same. First of all, the terms "regular" and "nonsingular" are synonyms, see Properties, Definition 02IS. Being normal and regular are the same thing for Noetherian 1-dimensional schemes (Properties, Lemmas 0569 and 0BX2). See Varieties, Lemma 0B8Y for the case of curves. Thus (3) is the same as (5). Finally, (6) is the same as (3) by Varieties, Lemma 0A26.

If $f : X \to Y$ is a nonconstant morphism of nonsingular projective curves, then $f$ sends the generic point $\eta$ of $X$ to the generic point $\xi$ of $Y$. Hence we obtain a morphism $k(Y) = \mathcal{O}_{Y,\xi} \to \mathcal{O}_{X,\eta} = k(X)$ in the category (1). If two morphisms $f, g : X \to Y$ gives the same morphism $k(Y) \to k(X)$, then by the equivalence between (1) and (2), $f$ and $g$ are equivalent as rational maps, so $f = g$ by Lemma 0BXZ. Conversely, suppose that we have a map $k(Y) \to k(X)$ in the category (1). Then we obtain a morphism $U \to Y$ for some nonempty open $U \subset X$. By Lemma 0BXY this extends to all of $X$ and we obtain a morphism in the category (5). Thus we see that there is a fully faithful functor (5)$\to$(1).

To finish the proof we have to show that every $K/k$ in (1) is the function field of a normal projective curve. We already know that $K = k(X)$ for some curve $X$. After replacing $X$ by its normalization (which is a variety birational to $X$) we may assume $X$ is normal (Varieties, Lemma 0BXR). Then we choose $X \to \overline{X}$ with $\overline{X} \setminus X = \{x_1, \ldots, x_n\}$ as in Varieties, Lemma 0BXW. Since $X$ is normal and since each of the local rings $\mathcal{O}_{\overline{X},x_i}$ is normal we conclude that $\overline{X}$ is a normal projective curve as desired. (Remark: We can also first compactify using Varieties, Lemma 0BXV and then normalize using Varieties, Lemma 0BXR. Doing it this way we avoid using the somewhat tricky Morphisms, Lemma 0BXB.) □